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Article

Rapid Analysis of Sugarcane with High Extraneous Matter by Near-Infrared Spectroscopy

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ABSTRACT: Efficient analysis of sugar cane quality is critical for the global sugar industry, which is valued at approximately $78 billion annually. Near-infrared (NIR) spectroscopy is a rapid, nondestructive alternative for analyzing multiple quality parameters of sugar cane. However, sugar cane with large amounts of extraneous matter (EM)—soil, green and brown leaves, roots, and tops—presents a unique challenge to NIR analysis due to the increased scattering and absorption from the EM, which is becoming increasingly relevant as climate change intensifies rainfall patterns, mechanical harvesting replaces manual methods, and environmental regulations promote green cane harvesting. This study evaluates the potential of NIR spectroscopy as a rapid, nondestructive alternative for analyzing multiple quality parameters of sugar cane with high EM values. Samples of shredded cane, pressed cane, and pressed juice were collected from three Louisiana (LA) sugar factories. Partial least squares regression models with k-fold cross-validation were developed to relate NIR spectra to reference values. The models demonstrated that pressed juice and pressed cane samples yielded the most accurate predictions of Brix, pol, and moisture, outperforming shredded cane samples. These findings support the practical implementation of NIR spectroscopy for the rapid and accurate determination of multiple quality parameters, from sugar cane with high EM.

KEYWORDS: *Near-Infrared Spectroscopy, Sugar Cane Quality, Extraneous Matter, Partial Least Squares Regression, Brix and Pol Prediction*

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■ INTRODUCTION

Enhancing agricultural productivity while simultaneously improving food safety and quality is a key challenge in the 21st century.[1] A technology that has played a vital role in addressing this challenge is near infrared (NIR) spectroscopy, a type of high-energy vibrational spectroscopy performed between 4000 and 13300 cm^{-1}.[2] NIR spectroscopy provides significant advantages over traditional wet chemistry methods for product characterization. It is nondestructive, environmentally friendly (as it uses no chemical reagents), and cost-effective over its lifespan, while also enabling rapid, *in situ* analysis of multiple parameters.[3] In addition, NIR also has advantages over other vibration techniques like mid-infrared (MIR) spectroscopy, since it features weaker water band intensities and smaller absorption coefficients,[4] allowing for better light penetration in high-moisture samples. Moreover, NIR requires minimal sample preparation to optimize the sample/light interaction,[5] making it ideal for analyzing heterogeneous materials such as sugar cane.

The sugar industry in certain countries has utilized NIR technology for nearly 30 years to characterize various materials involved in sugar production, including juices, syrups, massecuites, molasses, and raw sugar.[6−12] Characterizing the raw material, sugar cane, is especially critical for the cane payment system. In many sugar cane-producing regions, the land and crops often do not belong to the factories; thus, a fair payment system based on both the quantity and quality of the cane is essential. Most common payment systems measure key quality parameters such as pol (sucrose content by polarimetry), Brix (refractometric dissolved solids), and fiber. While these assessments are critical for fair compensation and maximizing processing efficiency, NIR models for cane characterization are often geographically specific. Sugar cane varieties, along with environmental and agricultural conditions, such as climate, soil type, and harvesting practices, vary significantly between regions, directly affecting sugar cane quality including levels of EM. Climate change has intensified rainfall patterns in some regions,[13,14] increasing EM specially soil content in harvested cane.[15,16] Additionally, the shift from manual harvesting to mechanical harvesting driven by the need to reduce production costs and address labor shortages, as well as the transition from harvesting burnt cane to green cane in compliance with environmental regulations has further contributed to rising EM levels in sugar cane-producing countries.[17,18] These factors make it challenging to transfer NIR calibration models from one region to another, as the models need to be tailored to the specific conditions and varieties present in each area.

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Louisiana (LA), for example, faces unique conditions that contribute to some of the highest levels of EM in sugar cane. The LA processing season is one of the shortest in the world, forcing factories to harvest and process cane at high rates under harsh environmental conditions, including freezes, rain, and hurricanes. These conditions lead to increased EM levels, significantly affecting cane quality and complicating the transfer of NIR calibrations from other sugar cane industries, even when the same instruments are used. Recent studies by Eggleston et al. reported EM levels in LA of up to 23.2 ± 2.1%,[19] much higher than those typically found in other cane-producing countries, such as Colombia (3.5−11.5% EM)[20] Sudan (3.7−9.5%),[21] and India (3.1−7.4%).[22] Notably, countries that have implemented NIR models for quality control and cane payment, like Australia and South Africa, reported significantly lower EM levels of 13−15%[23] and 5.7−11%,[24] respectively

Furthermore, the current payment system is additionally constrained by its labor-intensive and time-consuming nature, requiring teams of analysts to manually process each sample. A single sample can take up to 2−3 h to analyze, and not every truckload of cane can be tested, meaning growers are often paid based on a subset of their delivered cane, which can introduce laboratory errors and discrepancies, potentially leading to inaccurate payments. In this study, NIR spectroscopy was used to develop calibration models for predicting Brix, pol, and moisture in sugar cane with high EM levels, using samples from LA factories. These models address unique challenges of LA and offer insights for other regions facing increasing EM due to climate change, mechanization, and the transition from burnt to green harvesting.

■ MATERIALS AND METHODS

Samples. For the LA cane payment system, a sugar cane core sample from the truck is obtained, then it is shredded (shredded cane) and pressed (pressed cane) to extract the juice (pressed juice) and finally, the samples are analyzed in the laboratory. For this study, samples of shredded cane, pressed cane, and pressed juice were randomly collected directly from the factory laboratory at three LA sugar cane factories. The shredded cane and pressed cane samples were split into five subsamples, and NIR spectra of each subsample was collected. The five subsamples were mixed and then analyzed for pol and Brix (shredded cane) and moisture (pressed cane). The NIR spectra of the pressed juice were collected five times (thoroughly mixed before each collection) prior to pol and Brix analysis. A set of shredded cane and pressed cane samples were blended using a Ninja BL770 Blender (Needham, MA, USA) to reduce fiber size. All the samples were collected during the 2020 to 2023 LA processing seasons. A total of 1500 samples of shredded cane, 500 of pressed juice, and 400 of pressed cane samples were analyzed.

Laboratory Analysis. The Brix of pressed juice was measured using a digital Bellingham & Stanley RFM340 Refractometer (Tunbridge Wells, U.K.), with an accuracy of ±0.01 Brix. Before Brix analysis, samples were centrifuged at 3000 rpm for 10 min to remove interfering suspended particles. The Pol of pressed juice was measured at 589 nm using an Anton Paar MCP 5300 Sucromat polarimeter (Houston, TX, USA) at Factories 1 and 2, and an Optical Activity polAAr 30 polarimeter (Ramsey, U.K.) at Factory 3. Each juice sample was first clarified with Octapol clarification aid (Baddley Chemicals, Baton Rouge, LA, USA) and then filtered using Hoh PAK qualitative filter paper (New Iberia, LA, USA) placed in a 10.5 cm diameter funnel in a receiving beaker. For moisture analysis, a 50 g subsample of pressed cane was weighed and recorded to three decimal places, then placed in a preweighed aluminum boat. The sample was dried in an oven under three different conditions: (i) 105 °C for 25 h (ICUMSA Method GS7-5, 1994), (ii) 140 °C for 3 h, and (iii) 160 °C for 3 h (typical LA factory method). After the drying period, the dried sample in the aluminum boat was reweighed, and moisture content was calculated based on the mass difference of the sample before and after drying.

NIR Data. The NIR spectrometers used for the analysis were a MPA I FT-NIR spectrometer and a MPA II FT-NIR spectrometer (Bruker Scientific LLC, Billerica, MA, USA). For each sample, 128 scans were collected and averaged to produce a single spectrum. This process was repeated five times for each sample, with shredded cane and pressed cane samples split into five subsamples, and spectra collected for each. In contrast, for pressed juice, five spectra were collected from the same sample, and thoroughly mixed between each collection. The spectra were then averaged to develop calibration models. The measurement range was 3600 to 12480 cm^{-1} with a resolution of 16 cm^{-1}. A background spectrum was measured every 2 h. For shredded and pressed cane, the samples were analyzed by diffuse reflection using the 97 mm cup in the integrating sphere. For pressed juice, the samples were analyzed by transmission with the fiber optic probe.

NIR Calibration Models. Before calibration, the spectral data were preprocessed using either vector normalization or first derivative methods, chosen based on the lowest root mean square error (RMSE). The partial least squares (PLS) method was employed to correlate the variable matrix **Y** (pol, Brix, or moisture) with the spectral data. The predictive capacities of the NIR calibration models for pol and Brix in pressed juice were compared to those in shredded cane. For moisture content, the predictive capacities of the models in shredded cane were compared to those in pressed cane. Additionally, the impact of fiber size reduction, achieved by blending the shredded cane and pressed cane samples, on the prediction accuracy of the models was evaluated. Full cross-validation (leave-*p*-out) was applied during the development of the PLS model. Preprocessing and PLS modeling were conducted using OPUS Spectroscopy software, which is integrated with the Bruker FT-NIR spectrometer MPA I and MPA II.

Model Efficiency Estimation. To characterize the prediction capacity of calibration models, the coefficient of determination (R^2; eq 1) and root means square error of cross-validation (RMSECV; eq 2) were used.

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \tag{1}$$

$$\text{RMSECV} = \sqrt{\frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{N}} \tag{2}$$

where y_i and $\hat{y}_i$ are the measured value and the predicted value for sample i, respectively; $\bar{y}$ is the arithmetic mean of the measured data; N is the number of samples in a validation set.

■ RESULTS AND DISCUSSION

The raw NIR spectrum of 10 random pressed juice samples are shown in Figure 1a. The NIR spectra obtained after applying first derivate preprocessing are shown in Figure 1b. The spectra are shown only in the 7500−5500 cm^{-1} region since variable selection was also performed during the development of the calibration models to identify unique spectral features (wavelength regions) related to individual chemical components within a given sample.

Calibration models for predicting pol (Figure 2) and Brix (Figure 3) were developed for both pressed juice and shredded cane samples. To enhance the prediction accuracy for shredded cane, a set of samples was blended to reduce particle size. Previous studies have demonstrated that fiber size is a significant source of variation in the spectral profile and calibration statistics for forage[25] and maize silage.[26] Specifically, fiber sizes between 0.5 and 1.0 mm consistently improved

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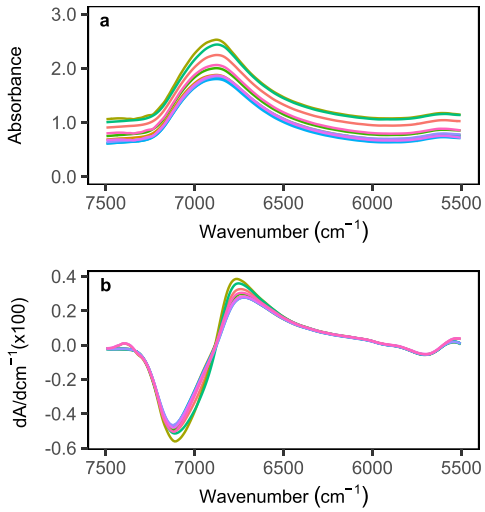

Figure 1. NIR spectra of pressed juice samples: (a) raw spectra; (b) first derivate preprocessing spectra for Brix calibration.

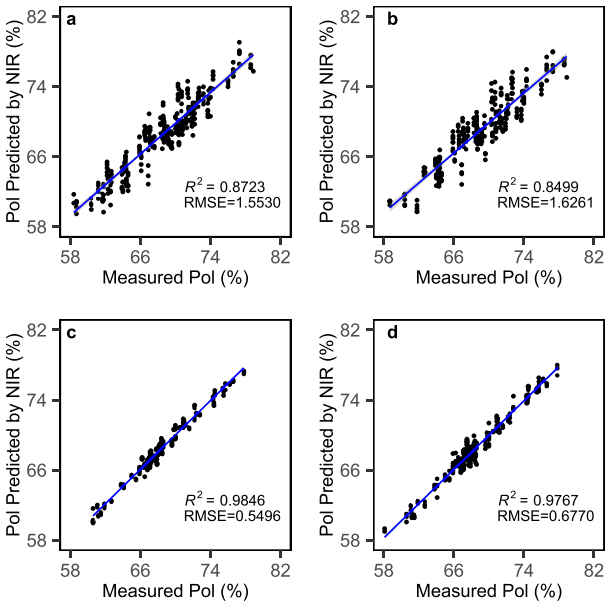

Figure 2. PLS calibration/cross-validation results for pol in (a) shredded cane, (b) blended shredded cane, (c) pressed juice with preamplifier gain A, and (d) pressed juice with preamplifier gain B.

calibration statistics.[25,27] In the current study, the fiber size of shredded cane ranged from 1 to 20 mm, depending on the shredder efficiency, and was reduced to a range of 1 to 5 mm after blending. For pressed juice, to improve the prediction accuracy of the NIR calibration models, the preamplifier gain—defined as the amount by which the input signal is amplified at the detector based on the strength of the signal—was adjusted. The preamplifier gain settings were evaluated at two different levels, A and B. Setting A multiplies the reference gain by 3, while setting B multiplies it by 10, allowing for adjustments tailored to different sample types. These settings were tested to determine the optimal amplification level for enhancing model accuracy.

Despite efforts to enhance sample homogeneity through fiber size reduction of shredded cane and adjusting the preamplifier gain for pressed juice, no significant improvement in calibration model prediction capacity was observed (Figure 2 and Figure 3). Comparing shredded cane to pressed juice revealed that the models for pol and Brix exhibited higher R^2 values and lower RMSE when pressed juice was analyzed. This outcome was anticipated since liquid samples like pressed juice are more homogeneous and do not involve the fiber size effect on the spectra.

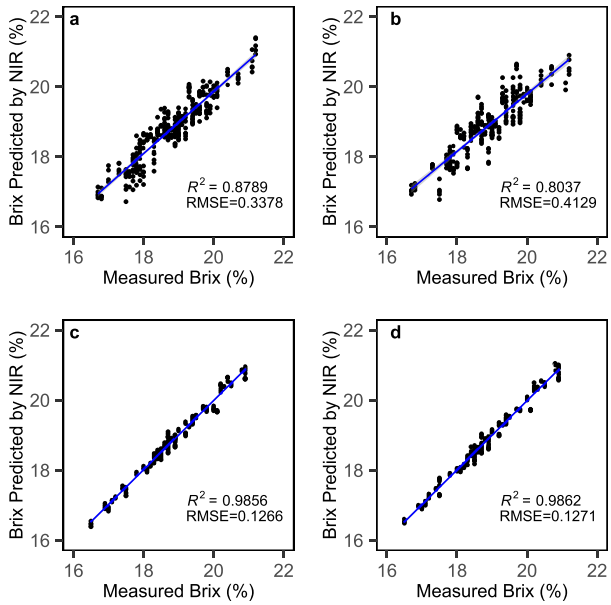

Figure 3. PLS calibration/cross-validation results for (a) Brix in shredded cane, (b) blended shredded cane, (c) pressed juice with preamplifier gain A, and (d) pressed juice with preamplifier gain B.

After determining that the highest performance in calibration models for pol and Brix was achieved when analyzing pressed juice, samples from three LA factories were analyzed, and calibration models were developed for each (Table 1). The performance of these models varied among factories and ultimately, the highest performance was achieved when combining the data from all three factories into a single calibration model.

To assess the precision of NIR calibration models compared to laboratory measurements for pol and Brix, two random samples of pressed juice were analyzed eight times using both methods. Statistical analysis revealed that for Sample 1, pol and Brix measurements were significantly different and exhibited greater variation with NIR, while for Sample 2, results from both methods were statistically comparable (Figure 4). These results demonstrate the potential of NIR spectroscopy as a promising tool for rapid analysis of pol and Brix; however, further refinement is necessary to enhance the predictive accuracy and consistency of the calibration models.

Calibration models for moisture were developed for shredded cane and pressed cane, and the effect of blending the samples to reduce fiber size was also evaluated. The calibrations for shredded cane were conducted using the moisture of the shredded cane measured at 60 °C for 48 h under vacuum. For pressed cane, two methodologies to determine moisture were compared: (i) drying at 105 °C for 25 h and (ii) drying at 160 °C for 3 h. The first drying temperature (105 °C) was chosen as it is recommended by the International Commission for Uniform Methods of Sugar Analysis (ICUMSA), and the second (160 °C) was selected because it is the most common temperature used in LA factories for drying pressed cane.

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Table 1. Performance of NIR Calibration Models for Brix and Pol in Pressed Juice

	Factory 1		Factory 2		Factory 3		All factories	
	R^2	RMSECV	R^2	RMSECV	R^2	RMSECV	R^2	RMSECV
Pol	96.5	1.010	93.6	1.080	97.1	1.050	97.2	1.120
Brix	98.5	0.128	97.8	0.121	97.8	0.179	98.6	0.156

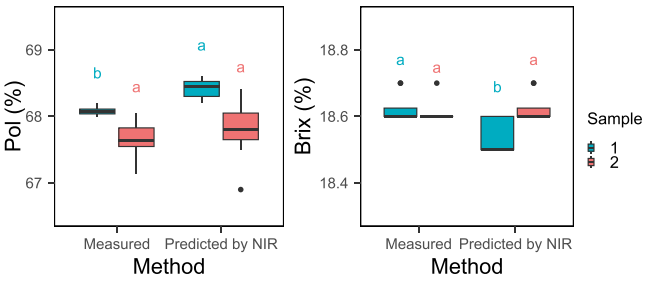

Figure 4. Precision of the NIR calibration models for Brix and pol in pressed juice. Boxplots show the median, interquartile range (IQR), and potential outliers. Different letters indicate significant differences at $p \leq 0.05$.

According to the results shown in Figure 5, blending the samples did not contribute to an improvement in the performance of the calibration models for either shredded cane or pressed cane in predicting moisture content. When comparing the calibration models for shredded cane versus pressed cane, the pressed cane models performed better, exhibiting higher R^2 values and lower RMSE. In terms of drying methodologies for pressed cane, the NIR calibration models based on moisture measured at 105 °C for 25 h performed better. This result was expected due to the lower accuracy of the moisture determination methodology at 160 °C for 3 h, which was confirmed after drying pressed cane samples at different temperatures (Figure 6). During these tests, it was observed that drying at higher temperatures (140 and 160 °C) increased the moisture content compared to lower temperatures (105 °C). Additionally, the samples dried at high temperatures turned brown, and the weight never became constant. These results strongly indicated that the apparent increase in moisture was not solely due to water loss but also the breakdown and evaporation of some organic components in the sample.

To assess the precision of NIR calibration models compared to laboratory measurements for moisture content, two random samples of pressed cane were analyzed using both methods. Statistical analysis revealed that the moisture content measured at 160 °C for 3 h was statistically higher than the moisture measured at 105 °C for 25 h and the moisture predicted by NIR (Figure 7). However, there was no statistical difference between the moisture measured at 105 °C for 25 h and the moisture predicted by NIR. This result confirms the success of the NIR calibration model for moisture, although there is room for improvement since the best R^2 was below 0.90 (Figure 7).

In conclusion, this study demonstrates that NIR spectroscopy is a powerful tool for accurately predicting key sugar cane quality parameters, including Brix, pol, and moisture in regions with high EM. The analysis of pressed juice for Brix and pol, as well as pressed cane for moisture, resulted in significantly better prediction accuracy compared to shredded cane samples. These findings highlight the potential of NIR spectroscopy to streamline sugar cane quality assessments, offering a rapid, nondestructive, and reliable alternative to

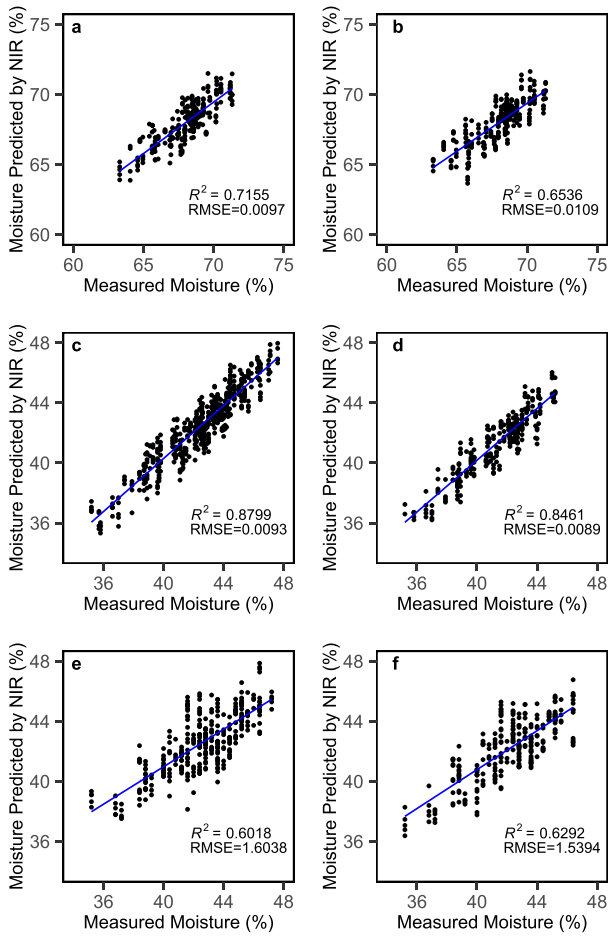

Figure 5. PLS calibration/cross-validation results for moisture content in (a) shredded cane dried at 60 °C for 48 h under vacuum, (b) blended shredded cane dried at 60 °C for 48 h under vacuum, (c) pressed cane dried at 105 °C for 25 h, (d) blended pressed cane dried at 105 °C for 25 h, (e) pressed cane dried at 160 °C for 3 h, and (f) blended pressed cane dried at 160 °C for 3 h.

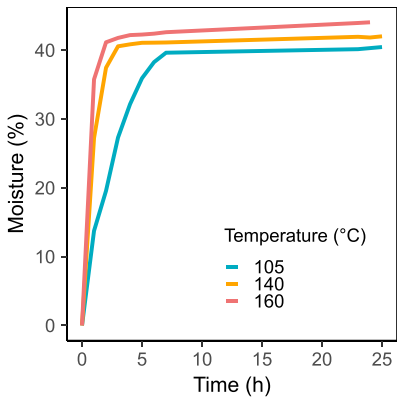

Figure 6. Effect of drying temperature and time on moisture content in pressed cane.

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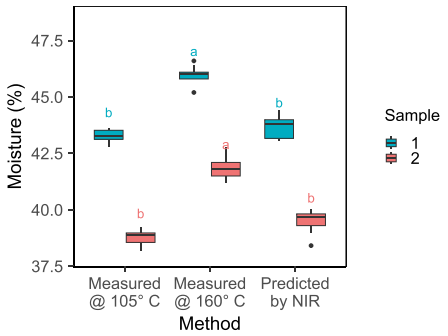

Figure 7. Precision of the NIR calibration models for moisture in pressed cane. Boxplots show the median, interquartile range (IQR), and potential outliers. Different letters indicate significant differences at $p \leq 0.05$.

traditional methods. It is important to note, however, that the accuracy of NIR calibrations is fundamentally dependent on the accuracy and precision of the wet chemistry methods used as primary reference standards. Therefore, ensuring the reliability of these primary wet chemistry methods is a critical first step in achieving robust and reliable NIR predictions. Future research should prioritize reviewing and refining these primary methods to improve the quality of NIR calibrations. Additionally, future research should focus on validating these predictive models with independent sample sets and investigating advanced, nonlinear calibration techniques to further enhance the accuracy and robustness of NIR calibration models. Overall, this research underscores the practical value of NIR spectroscopy in optimizing the sugar cane payment system, ultimately benefiting both growers and processors by enabling more precise and efficient quality evaluation.

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Notes

The authors declare no competing financial interest.

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